Rashmiprakash Satapathy

+91-7854989076 | rpsatapathy04@gmail.com | linkedin/rpsatapathy04 | github/Rashmiprakash-56

Professional Summary

Software Engineer with experience in building full-stack and backend systems using **React.js**, **Django**, **Node.js**, **Python**, and SQL/NoSQL databases. Skilled in **REST API** design, distributed and concurrent systems, database optimization, and **CI/CD**, with a strong foundation in **C++**, **Data Structures & Algorithms**, and **Object-Oriented Programming**.

Education

National Institute of Technology Raipur 2021-2025
Bachelor of Technology (CGPA: 8.36) *Raipur, Chhattisgarh*

Experience

Mercados Energy Markets India Limited | *Analyst (Software Development)* July 2025 – Present

- Developed and integrated **full-stack billing features** using **React.js/Redux** and **Django REST APIs** for a production bill verification platform serving **UPPCL**, supporting **300+ power plants**.
- Engineered plant-specific **bill calculation and data-processing logic** in **Django**, handling divergent per-plant formats and calculation rules and integrating results.
- Optimized **MongoDB** query performance and backend business logic over a **100K+ record** dataset using **bulk write strategies**, aggregation pipelines, and a modular service-layer architecture to ensure correctness at scale.
- Built a **Python optimization script** for resource adequacy planning, solving multi-constraint energy problems to support procurement cost analysis.

Mercados Energy Markets India Limited | *Intern* Dec 2024 – June 2025

- Built a **day-ahead electricity demand forecasting pipeline** via a custom **ETL pipeline** processing **10M+** SCADA, weather, and market records, cutting preprocessing time by **50%**; modeled with **XGBoost, Random Forest, and stacking ensembles** on engineered weather/temporal features, achieving under **3% MAPE** – a **12% improvement** over baseline.
- Delivered **Power BI dashboards** and automated Python analytics reports for demand monitoring, cutting manual reporting effort by **40%**.

Technical Skills

Languages: C++, Python, Java, SQL, JavaScript, Typescript
Core CS: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), System Design, DBMS
Backend & Frontend: Django, FastAPI, Node.js, Express.js, REST APIs, React.js, Redux, Tailwind CSS
Databases & Infrastructure: PostgreSQL, MySQL, MongoDB, Redis, Data Modeling, Query Optimization, Indexing, AWS, Docker, CI/CD, Git, GitHub, Agile
AI/ML: XGBoost, Scikit-learn, LangGraph, Vector Embeddings, Semantic Search, RAG, LLM Integration

Projects

Asynchronous Task Queue Engine | *Node.js, Express, PostgreSQL, Redis, Docker* [GitHub]

- Architected a horizontally-scalable, concurrent task queue in **Node.js**, **PostgreSQL**, and **Redis** with **priority-based job queues** and row-level locking for safe parallel job consumption across multiple workers.
- Engineered fault tolerance with a **dead-letter queue**, automated retries, a **Redis-backed circuit breaker**, and stale-job recovery; **containerized** all services with **Docker Compose** and exposed **Prometheus**-format metrics.

Multi-Agent Chat System | *React, FastAPI, MongoDB, LangGraph* [GitHub] [Live Site]

- Built a full-stack chat platform with a **supervisor-agent architecture** routing queries across **3 specialized sub-agents** (RAG, NL-to-SQL, Web Search), **real-time SSE token streaming**, and a Pinecone vector search pipeline.
- Secured the system with **JWT authentication** and persistent multi-turn memory via **PostgreSQL-backed checkpointers**, with end-to-end distributed tracing across agent/tool calls using LangSmith.

Time-Series Demand Forecasting Pipeline | *React, FastAPI, XGBoost, GitHub Actions* [GitHub] [Live Site]

- Built a full-stack app serving hourly electricity demand forecasts over a **35,000+ row** dataset, with **async background task processing** and real-time job-status polling for a responsive UI during training.
- Automated model versioning/deployment to **Hugging Face Hub** via **GitHub Actions CI/CD**; exposed **REST APIs** for 24-hour forecasts with MAPE evaluation and SHAP explainability dashboards.

Coding Profiles

LeetCode: **500+** questions solved | Contest Rating: **1927 (Knight)**
GeeksforGeeks: **400+** questions solved

Achievements

- Achieved **97.53 percentile** in JEE Main 2021 and **94.4%** in CBSE Class XII Board Examinations.
- Secured **1st Rank** in Electrical Engineering at **Vigyaan**, a National Level Science Exhibition organized by NIT Raipur.